

Comprehensive Digital Evidence Acquisition and Command-Line Analysis using FTK Imager

EXPERIMENT NO: 01

DATE:

1. AIM

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager, capturing live volatile memory (RAM) using the WinPmem command-line utility, and analyzing the extracted memory dump using Volatility 3 commands via the PassMark Volatility Workbench.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tools:**
 - AccessData FTK Imager (GUI for non-volatile physical drive imaging)
 - WinPmem (winpmem_mini_x64_rc2.exe) (CLI tool for volatile memory capture)
 - Volatility 3 Framework (vol.exe / PassMark Volatility Workbench V3.0)
- **Hardware:** Evidence PC/Workstation, Target Storage (sanDisk cruzer Blade USB Device [16GB USB])

3. DESCRIPTION

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.
- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).

- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., `-f <image_path> windows.pslist.PsList`) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

4. PROCEDURE

Step 1: Select the Source Type

Open FTK Imager as an Administrator. Navigate to **File > Create Disk Image**. In the Select Source dialog, choose **Physical Drive**.

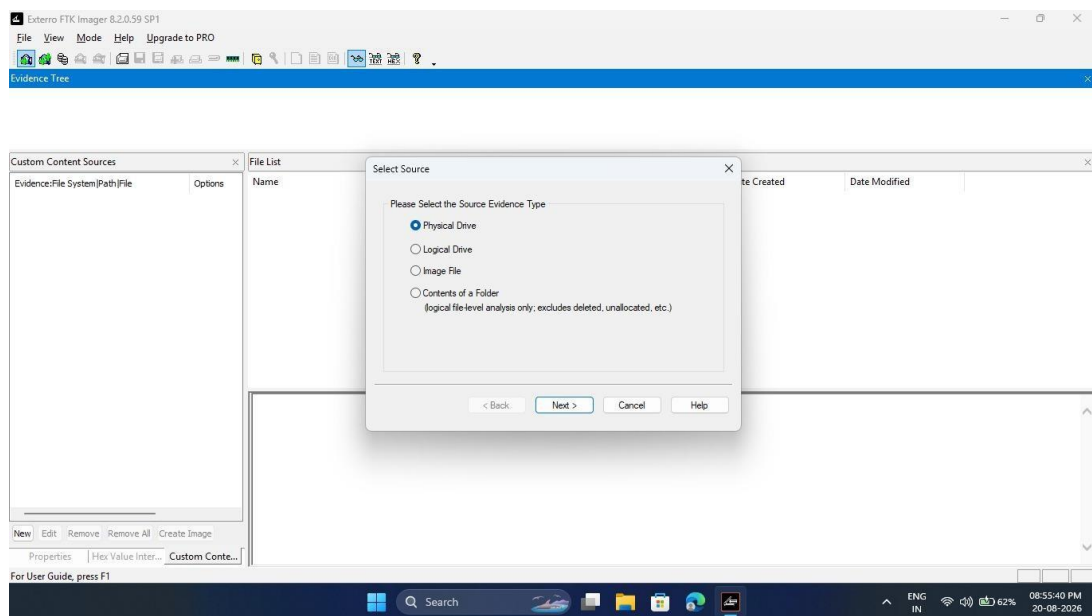


Figure 1: Selecting Physical Drive as the source evidence type

Step 2: Select the Specific Drive

Select the target evidence drive (\\.\PHYSICALDRIVE2 - sanDisk cruzer Blade USB Device [16GB USB]) from the dropdown.

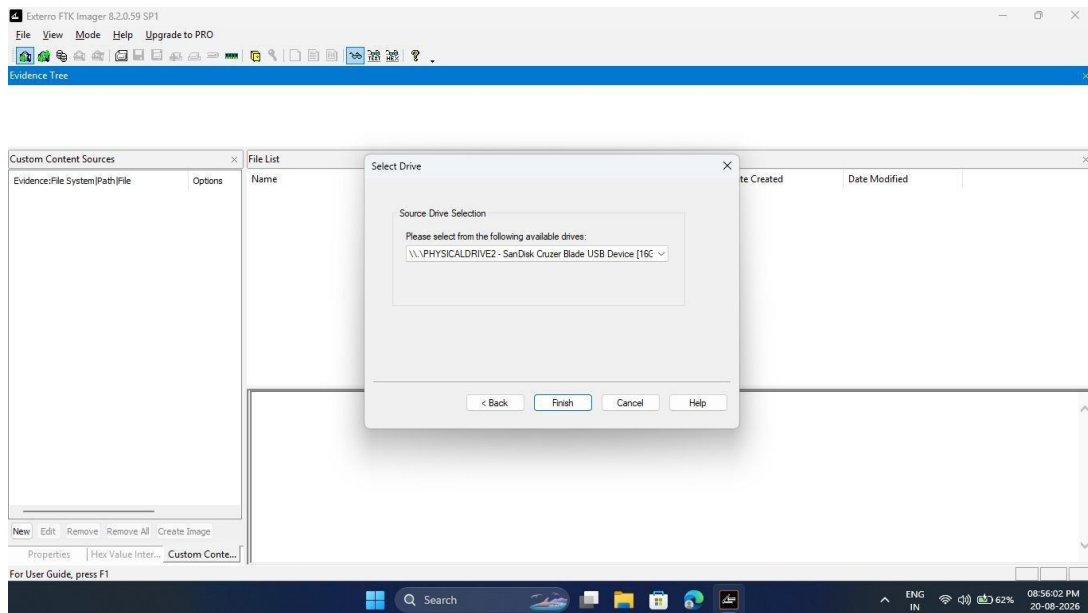


Figure 2: Selecting the target USB device for acquisition

Step 3: Choose the Image Type

Select E01 as the destination image type to create a pure bitstream copy.

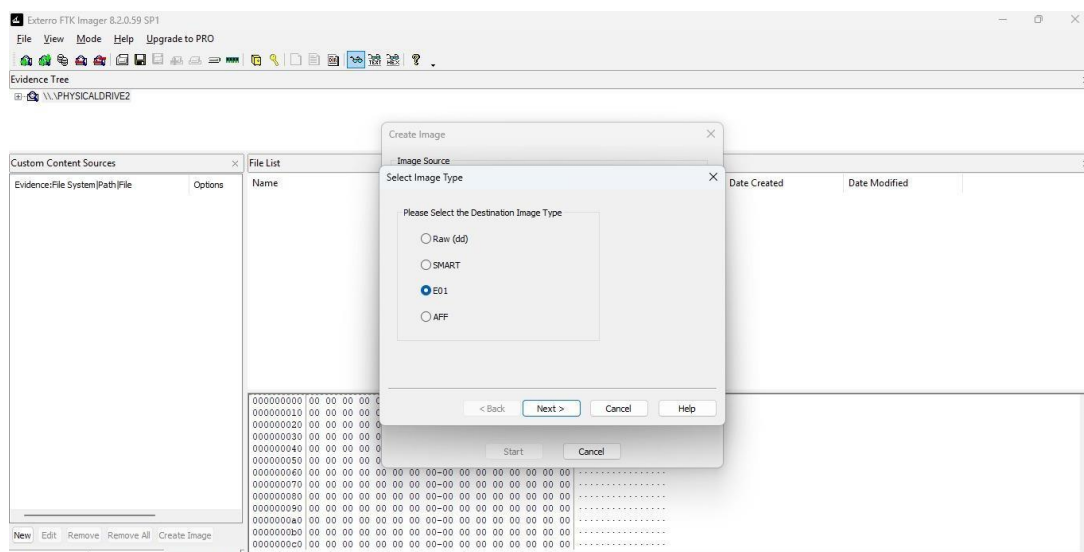


Figure 3: Configuring the destination image format to E01

Step 4: Enter Evidence Item Information

Provide the standardized case details to maintain the chain of custody:

- **Case Number:** CASE001
- **Evidence Number:** 1
- **Unique Description:** USB Pen Drive
- **Examiner:** Suryaprakash T

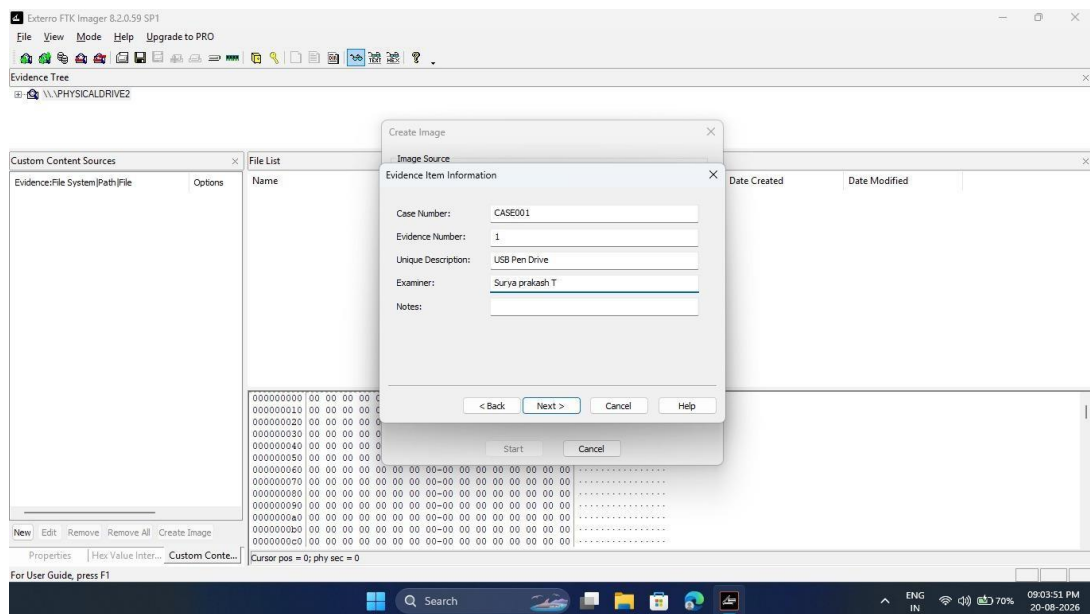


Figure 4: Entering the forensic case and examiner details

Step 5: Configure Image Destination

Set the destination folder to A:\DF\Digital_Forensics_Lab\Experiment-01-FTK-Imager\Results and the filename to Result. Set the Fragment Size to 1500 MB and check **Verify images after they are created**.

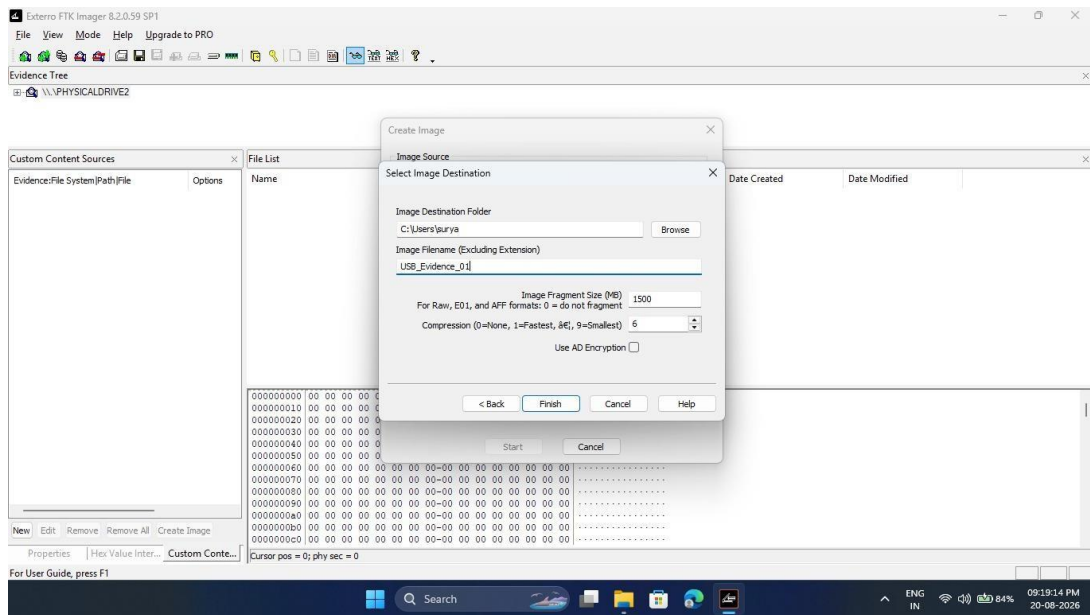


Figure 5: Setting the destination path, filename, and fragmentation size

Step 6: Monitor Acquisition Progress & Hash Verification

Click **Start**. Once complete, FTK Imager hashes the original drive and the newly created image. Review the Drive/Image Verify Results.

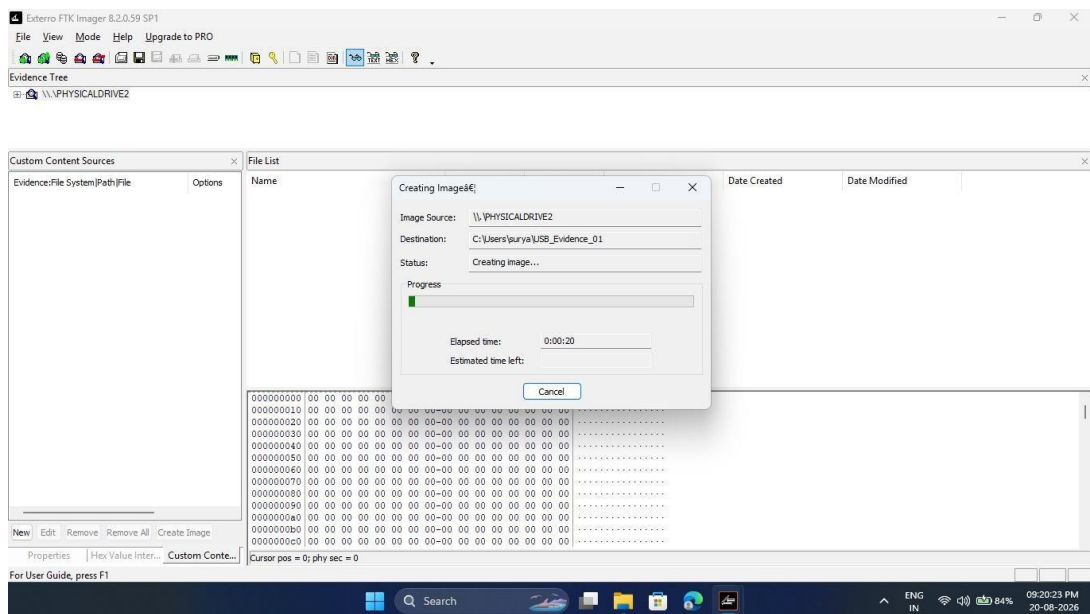


Figure 6: Disk imaging progress tracking

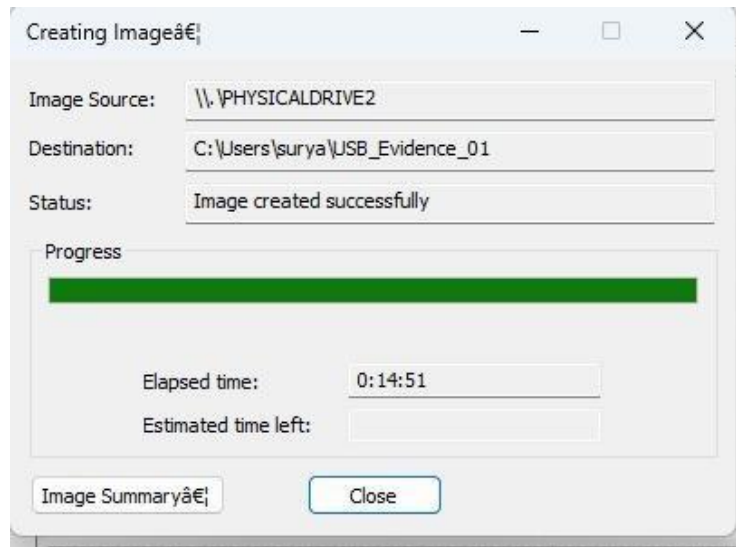


Figure 7: Successful verification of MD5 and SHA1 hashes

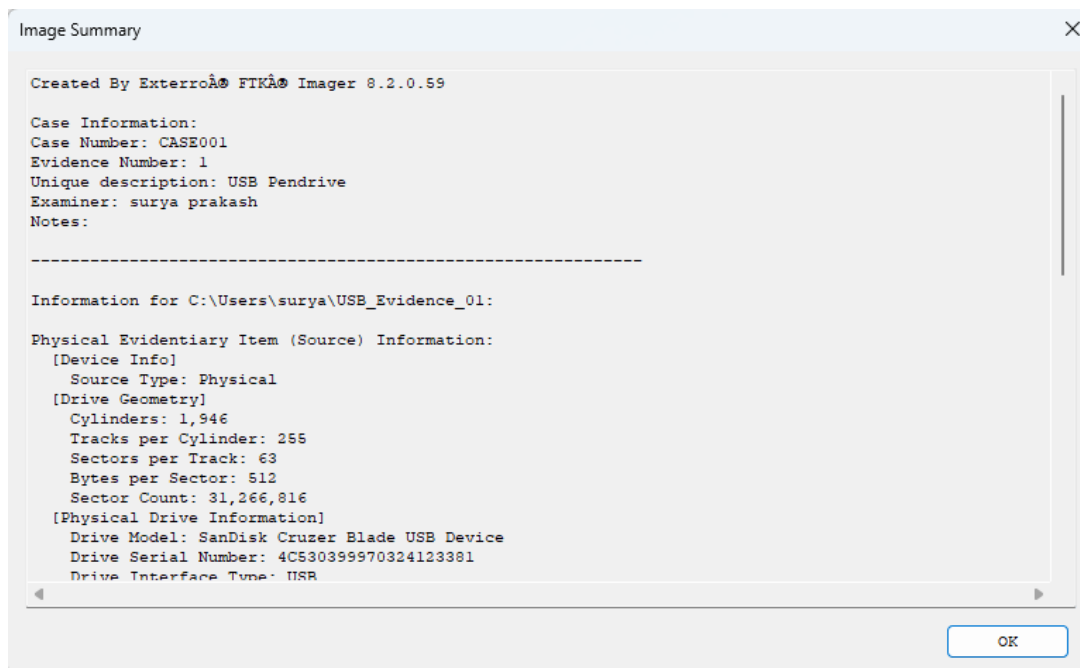


Figure 8: Image Summary detailing the physical device geometry

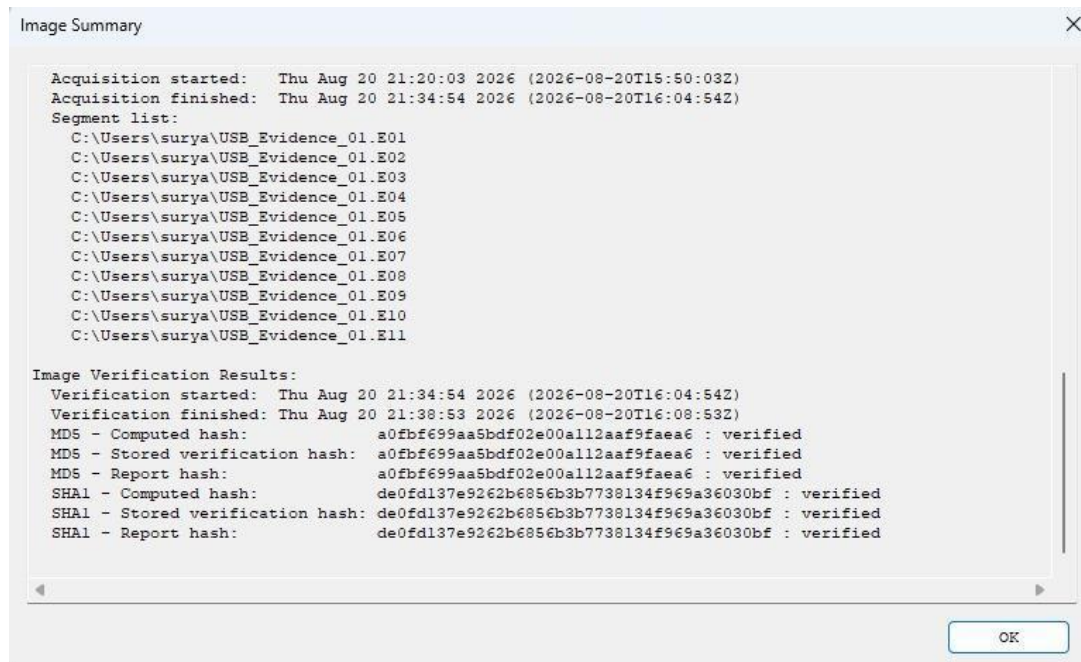


Figure 9: Image Summary detailing the hash verification results and segment list

6. RESULT

Conclusion : The digital evidence acquisition process was successfully completed using FTK Imager. A bit-by-bit E01 image of the target 16GB USB storage device was created while preserving the integrity of the original evidence. The evidence details and case information were properly recorded to maintain the chain of custody. The acquired image was successfully verified using MD5 and SHA1 hash values, confirming the integrity and authenticity of the forensic image. Thus, the experiment demonstrated the proper procedure for acquiring and verifying non-volatile digital evidence in a forensic investigation.